

**VIETNAM GENERAL CONFEDERATION OF LABOUR
TON DUC THANG UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY**



**BUI BACH KHANH DUY
TRAN THIEU KHANG**

**DESIGN AND DEVELOPMENT OF A
VIETNAMESE FINANCIAL CHATBOT
INTEGRATING A MULTIMODAL PRICE
MOVEMENT PREDICTION MODEL**

**UNDERGRADUATE THESIS
OF
COMPUTER SCIENCE**

HO CHI MINH CITY, YEAR 2026

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Advised by
M.Sc. Le Anh Cuong

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DECLARATION OF AUTHORSHIP

We hereby declare that this thesis was carried out by ourself under the guidance and supervision of M.Sc. Le Anh Cuong; **and that the work and the results contained in it are original** and have not been submitted anywhere for any previous purposes. The data and figures presented in this thesis are for analysis, comments, and evaluations from various resources by our own work and have been duly acknowledged in the reference part.

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ABSTRACT

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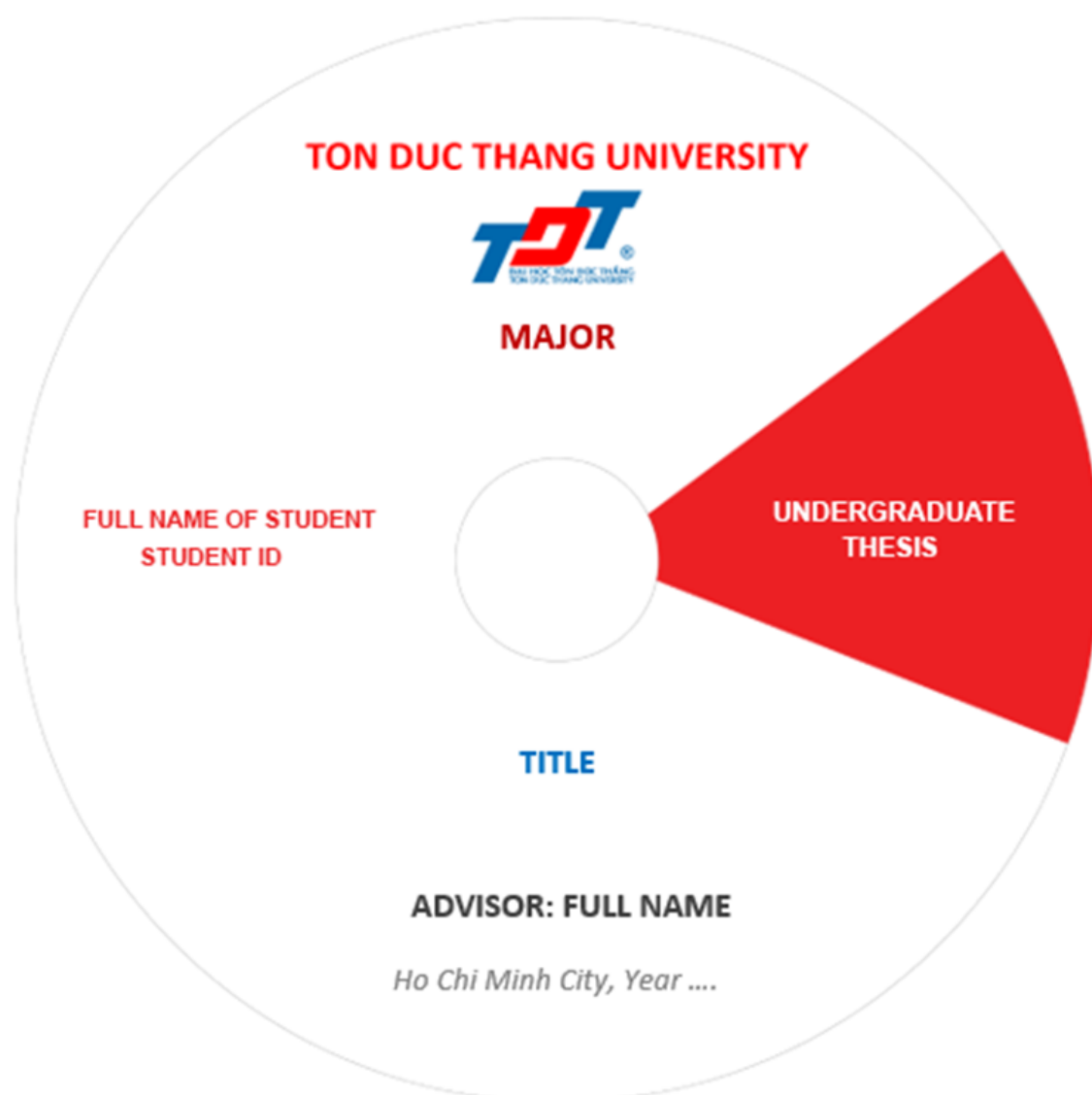
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ABBREVIATIONS

ACM	Association for Computing Machinery
AD	Average Drop
AG	Average Gain
AI	Average Increase
BMVC	British Machine Vision Conference
CAM	Class Activation Mapping
CNN	Convolutional Neural Network
CVPR	IEEE Conference on Computer Vision and Pattern Recognition
ECCV	European Conference on Computer Vision
FP16	16-bit Floating Point
GPU	Graphics Processing Unit
GradCAM	Gradient-weighted Class Activation Mapping
ICCV	IEEE International Conference on Computer Vision
ICLR	International Conference on Learning Representations
IEEE	Institute of Electrical and Electronics Engineers
KD	Knowledge Distillation
NeurIPS	Neural Information Processing Systems
OptiCAM	Optimized Class Activation Mapping
ResNet	Residual Network
RGB	Red Green Blue
TDTU	Ton Duc Thang University
WACV	IEEE Winter Conference on Applications of Computer Vision
XAI	Explainable Artificial Intelligence



CHAPTER 1 - INTRODUCTION

The intersection of artificial intelligence and quantitative finance has catalyzed a paradigm shift in how market participants process information, assess risk, and execute trading strategies. Traditional financial forecasting approaches, including econometric models and statistical learning methods, often rely on structured numerical inputs and simplifying assumptions about market efficiency. However, such approaches struggle to capture the complexity of modern financial markets, where unstructured data sources—such as news, reports, and textual disclosures—play a critical role in influencing price dynamics. This limitation has driven the emergence of multimodal machine learning frameworks capable of jointly modeling time-series data and natural language, enabling a more comprehensive representation of market behavior [1, 2].

Within this global evolution, the Vietnamese stock market represents a particularly compelling case study for the deployment of advanced financial AI systems. As a rapidly developing market transitioning toward emerging market status, Vietnam has experienced significant growth in retail investor participation. The structural composition of the market is heavily retail-driven, leading to high liquidity but also increased volatility and susceptibility to behavioral biases. Empirical studies have identified the presence of herding behavior and sentiment-driven trading patterns, which weaken the explanatory power of traditional financial theories such as the risk-return relationship [3, 4]. Additionally, macroeconomic conditions and external capital flows further amplify market sensitivity, reinforcing the need for adaptive and data-driven analytical approaches [5–7].

Despite the increasing complexity of the market, existing financial advisory tools in Vietnam remain largely deterministic and limited in analytical depth. Most platforms rely on predefined indicators and rule-based systems, lacking the capability to integrate heterogeneous data sources or generate context-aware insights. In parallel, recent advances in Vietnamese natural language processing have significantly improved the feasibility of building intelligent financial systems tailored to the local language. The development of models such as PhoBERT, PhoGPT, and VinaLLaMA has enabled more accurate representation and understanding of Vietnamese financial text, addressing previous linguistic bottlenecks in NLP applications [8–11].

In response to these challenges, this research proposes the design and development of an intelligent financial chatbot system based on a **multi-agent architecture** tailored to the Vietnamese stock market. The system integrates a multimodal predictive model with a set of specialized agents responsible for distinct tasks, including market data processing, financial news sentiment analysis, and contextual reasoning. These agents interact through coordinated communication and task allocation mechanisms, enabling the system to gener-

ate structured, context-aware, and explainable insights. This multi-agent approach enhances system flexibility, reduces reliance on a single model, and improves robustness through cross-agent verification and collaborative reasoning [1, 12].

The primary objectives of this research are structured across three key dimensions. First, to develop a multimodal learning framework that effectively fuses time-series financial data with textual sentiment signals derived from Vietnamese language models, leveraging advances in sequence modeling architectures such as transformers and temporal fusion techniques [12, 13]. Second, to design and implement a multi-agent architecture in which individual agents perform specialized roles such as data acquisition, prediction, analysis, and explanation, thereby enhancing system reliability and interpretability through coordinated reasoning. Third, to evaluate the proposed system within a simulated financial environment, assessing its ability to improve predictive accuracy and reduce the impact of behavioral biases in retail investment decision-making [2].

Ultimately, this research aims to contribute to the democratization of advanced financial analytics in Vietnam by providing an accessible, intelligent, and trustworthy decision-support system. By integrating multimodal machine learning, domain-specific language models, and a collaborative multi-agent framework, the proposed approach seeks to enhance transparency, reduce informational asymmetry, and support more rational investment behavior in a retail-dominated market.

CHAPTER 2 - BACKGROUND

To establish a rigorous theoretical and technical foundation for the design of a multimodal financial chatbot, it is essential to analyze the structural characteristics of the Vietnamese market, the evolution of language models tailored to this linguistic context, and the computational paradigms enabling real-time, data-grounded reasoning.

The Vietnamese Equity Market: Structure, Constraints, and Volatility

The Vietnamese stock market operates as a regulated ecosystem comprising the Ho Chi Minh City Stock Exchange (HOSE), the Hanoi Stock Exchange (HNX), and the UP-CoM market. Each exchange serves distinct roles across large-cap, mid-cap, and transitional equities. Market operations are governed by strict regulatory frameworks, including T+2 settlement cycles, pre-funding requirements, and daily price limits designed to control volatility [14–16]. These constraints significantly influence the design of predictive models, requiring a focus on multi-day forecasting horizons rather than high-frequency trading strategies.

Behaviorally, the market is dominated by retail investors, leading to pronounced inefficiencies. Empirical studies confirm the presence of herding behavior and momentum-driven trading, particularly during volatile periods, which weakens the applicability of traditional financial theories [3, 4]. Additionally, macroeconomic factors such as foreign direct investment and global economic conditions further impact market dynamics, reinforcing the need for models capable of integrating both structured and unstructured information sources [5–7].

Foundations of Vietnamese Natural Language Processing

The effectiveness of a financial chatbot depends critically on its ability to process Vietnamese text accurately. Unlike Western languages, Vietnamese exhibits unique linguistic characteristics, including syllable-based tokenization and complex diacritics, which pose challenges for standard NLP pipelines. Early solutions relied on preprocessing tools such as VnCoreNLP to perform word segmentation and syntactic analysis [11].

The introduction of pre-trained Vietnamese language models marked a significant advancement. PhoBERT established strong performance in downstream NLP tasks through large-scale pretraining on Vietnamese corpora [8]. Subsequent generative models, such as PhoGPT and VinaLLaMA, extended these capabilities to instruction-following and conversational reasoning, enabling more natural and context-aware text generation [9, 10].

More recently, domain-specific models such as MoFin have addressed the need for precise numerical reasoning in financial contexts. By integrating structured reasoning and

executable operations, these models bridge the gap between natural language understanding and quantitative computation, a critical requirement for financial applications [17].

Multi-Agent Architectures for Financial Reasoning

A fundamental limitation of large language models in financial applications lies in their reliance on static training corpora and their susceptibility to hallucination effects [18]. To address these challenges, recent research has increasingly focused on **multi-agent architectures**, where multiple autonomous agents collaborate to perform complex reasoning tasks.

In a multi-agent system (MAS), each agent is designed to specialize in a particular function, such as data retrieval, sentiment analysis, forecasting, or decision explanation. Through structured communication protocols and iterative interaction, these agents are able to decompose complex financial queries into smaller sub-tasks and produce more reliable and interpretable outputs. Compared to monolithic models, this distributed approach improves flexibility, robustness, and scalability, while enabling cross-agent verification that enhances consistency in high-stakes financial environments [19, 20].

Recent studies further highlight that LLM-based multi-agent systems extend classical MAS frameworks by incorporating semantic reasoning, planning, and adaptive coordination mechanisms. This evolution allows agents not only to exchange low-level signals but also to collaborate at the level of abstract knowledge and decision-making processes, making them particularly suitable for financial decision-support systems where transparency and interpretability are critical [19].

Pretrained Models for Financial Time-Series Forecasting

Financial time-series forecasting presents significant challenges due to noise, non-stationarity, and limited labeled data. To overcome these limitations, recent advances have introduced **pretrained time-series foundation models** that leverage large-scale datasets to learn transferable temporal representations.

A prominent example is the Chronos framework, which adapts transformer-based language models to time-series forecasting by tokenizing numerical sequences into discrete representations and training them using a cross-entropy objective [21]. Chronos models, pretrained on diverse datasets including synthetic and real-world time series, demonstrate strong zero-shot forecasting capabilities and competitive performance compared to models trained specifically for individual tasks. This approach illustrates how transfer learning can significantly improve generalization in financial prediction scenarios.

Beyond foundation models, hybrid deep learning architectures combining Long Short-Term Memory (LSTM) networks and Transformers have also shown strong performance in financial forecasting tasks. While LSTM networks effectively capture local temporal dependencies, Transformer models excel at modeling long-range interactions through self-attention mechanisms. Empirical studies demonstrate that combining these architectures leads to improved robustness and predictive accuracy in stock price forecasting and volatility modeling tasks [22, 23]. These developments highlight the growing importance of pretrained and hybrid architectures in building reliable predictive systems for financial markets.

Multimodal Machine Learning in Financial Time Series

Financial markets are inherently multimodal, with price movements influenced by both historical time-series data and external textual information. Traditional unimodal models, such as LSTM networks, capture temporal dependencies but fail to incorporate semantic context [24]. Multimodal machine learning addresses this limitation by integrating heterogeneous data sources into a unified predictive framework [1].

Fusion strategies are typically categorized into early, late, and hybrid approaches. Early fusion combines raw features before model input, while late fusion aggregates independent model outputs. Hybrid fusion, which operates at the feature level using shared latent representations, has emerged as the most effective approach. By leveraging cross-attention mechanisms, hybrid models can dynamically capture interactions between modalities, leading to significant improvements in predictive accuracy [1, 25].

Empirical evidence demonstrates that multimodal approaches outperform unimodal baselines in financial forecasting tasks, particularly in environments characterized by high volatility and strong sentiment influence. These models are therefore well-suited for the Vietnamese market, where price dynamics are closely tied to news and investor behavior.

CHAPTER 3 - RELATED WORK

The development of a multimodal financial chatbot tailored for the Vietnamese market lies at the intersection of several research domains, including financial sentiment analysis, time-series forecasting, and conversational AI systems. These fields have experienced rapid advancements due to deep learning and foundation models; however, their integration within emerging markets such as Vietnam remains relatively limited. This chapter reviews existing studies and highlights critical gaps that motivate the proposed approach.

Financial Sentiment Analysis in the Vietnamese Context

Financial sentiment analysis has been widely recognized as a significant factor in explaining stock price movements and market volatility. Empirical studies demonstrate that incorporating news sentiment into volatility models can significantly enhance predictive performance. For instance, sentiment-driven GARCH models show that both contemporaneous and lagged news sentiment contribute to volatility, with negative sentiment exhibiting a stronger and more persistent effect than positive sentiment [26].

Recent studies using transformer-based NLP models such as BERT further confirm that negative sentiment is strongly associated with increased market volatility and reduced market stability, particularly during geopolitical or economic uncertainty [27]. These findings highlight the asymmetric role of sentiment and validate its importance as a predictive signal in financial markets.

In the Vietnamese context, applying these techniques presents additional challenges due to linguistic complexity and limited domain-specific datasets. The introduction of PhoBERT has significantly improved Vietnamese NLP performance by leveraging large-scale monolingual corpora [8]. Subsequent datasets such as ViFinClass provide domain-specific benchmarks for financial sentiment classification, enabling more robust evaluation. Empirical evidence consistently shows that monolingual Vietnamese models outperform multilingual counterparts in domain-specific tasks, emphasizing the importance of localized language modeling.

The importance of sentiment is further amplified by the structure of the Vietnamese stock market. Retail investors dominate trading activity, accounting for approximately 90% of daily trading value, while the number of domestic investor accounts exceeded 12 million in 2026 [28, 29]. This retail-driven structure increases market sensitivity to news and behavioral biases, reinforcing the need to incorporate sentiment analysis into predictive frameworks.

Multimodal Stock Price Prediction Architectures

Stock price prediction is inherently challenging due to the noisy, non-linear, and non-stationary nature of financial time series. Traditional statistical models such as ARIMA and GARCH are limited in their ability to capture complex temporal dependencies and external influences. Deep learning models, particularly Long Short-Term Memory (LSTM) networks, have been widely adopted due to their ability to model sequential data [24].

However, LSTM models often struggle with long-range dependencies, leading to the adoption of transformer-based architectures. Transformers leverage self-attention mechanisms to capture global temporal relationships and have demonstrated superior performance in financial forecasting tasks [12]. Hybrid architectures combining CNNs and Transformers have shown improved predictive accuracy in stock movement classification tasks, outperforming traditional models in benchmark datasets [22].

Recent research has increasingly focused on multimodal learning, which integrates heterogeneous data sources such as time-series data, financial news, and chart-based representations. Multimodal fusion frameworks enable models to capture interactions between structured numerical data and unstructured textual information. Empirical studies show that multimodal approaches significantly improve forecasting accuracy and robustness, particularly in volatile markets influenced by sentiment [1, 30].

Furthermore, benchmark datasets such as FinMME, which includes over 11,000 multimodal financial samples, demonstrate that even state-of-the-art models struggle with complex financial reasoning tasks, highlighting the need for more advanced architectures that can effectively integrate and reason across modalities [31].

Conversational AI and Financial Assistants

The advancement of large language models has enabled the development of conversational AI systems for financial applications. Domain-specific models such as BloombergGPT, with over 50 billion parameters and trained on approximately 709 billion tokens, demonstrate strong performance in financial text understanding and reasoning tasks [32]. Open-source frameworks such as FinGPT further democratize financial AI by enabling efficient fine-tuning on domain-specific datasets [33].

In Vietnam, the application of conversational AI has been primarily concentrated in the banking sector. Major commercial banks, including Vietcombank and VietinBank, have deployed AI-powered chatbots to automate customer service and improve operational efficiency [34]. For example, Vietcombank's VCB Digibot handled over 2 million interactions within its first six months of deployment, with approximately 88% of user queries processed automatically [35].

These systems provide 24/7 support, automate routine queries, and reduce operational costs. However, most deployed solutions are based on rule-based or narrowly trained AI models that focus on predefined tasks such as account inquiries and transaction support. They lack advanced reasoning, multimodal integration, and predictive capabilities required for financial analysis [34].

In contrast, enterprise-level financial AI systems such as LV Financial AI Agent focus on analyzing internal financial statements and supporting corporate decision-making based on structured accounting data. While these systems provide real-time dashboards and trend analysis, they do not operate on market-wide data or support stock investment decisions, highlighting a clear distinction between enterprise financial analytics and market prediction systems.

Identification of the Research Gap

A comprehensive review of the literature reveals several critical gaps. First, although Vietnamese NLP models such as PhoBERT and VinaLLaMA achieve strong performance in language tasks, their integration into multimodal financial prediction systems remains underexplored. Second, existing predictive models often provide numerical outputs without sufficient interpretability, limiting their usability for retail investors.

Third, conversational AI systems in Vietnam are predominantly designed for customer service and lack advanced reasoning capabilities necessary for financial decision support. These systems do not integrate multimodal data sources or provide forward-looking predictions. Fourth, existing enterprise financial AI systems focus on internal data analysis rather than market-wide forecasting, leaving a significant gap in stock investment support.

Finally, despite strong empirical evidence of sentiment-driven volatility and behavioral biases in retail-dominated markets, there is a lack of systems that explicitly incorporate these factors into predictive models.

To address these limitations, this research proposes a unified framework that combines multimodal learning, Vietnamese sentiment analysis, and advanced reasoning mechanisms within a conversational AI system. The objective is to develop an intelligent financial assistant capable of delivering accurate, interpretable, and context-aware insights tailored to the Vietnamese stock market.

CHAPTER 4 - METHODOLOGY

Methodology Overview

This chapter details the overarching methodology employed to address the primary research objective: the *Design and development of a Vietnamese financial chatbot integrating a multimodal price movement prediction*. To capture the volatile, noisy, and multifaceted nature of financial markets, this research adopts a mixed-methods paradigm that integrates rigorous quantitative predictive modeling with experimental systems engineering [36]. Rather than relying solely on theoretical propositions, this study proposes a novel, comprehensive system and framework designed from the ground up. This approach is highly suitable for the problem domain because predicting stock movements and delivering actionable insights requires not only the sophisticated fusion of heterogeneous data streams—such as quantitative historical prices and unstructured textual news—but also robust validation mechanisms and an accessible deployment medium for end-users. To systematically achieve this, the research methodology is structured into three distinct but sequential phases: the construction of a multimodal prediction engine, the expansion into a multi-agent validation architecture, and the final deployment as an interactive chatbot service.

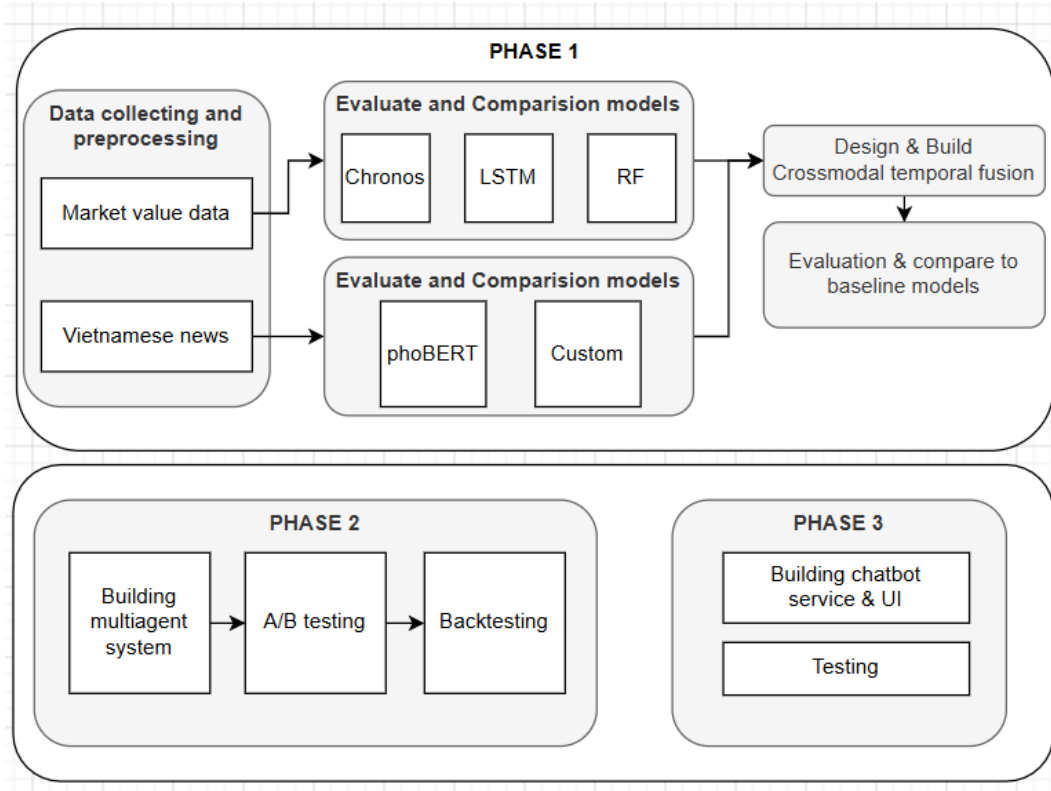


Figure 4.1: Overall methodology framework for the proposed multimodal financial chatbot system.

The first phase of the research is strictly experimental and quantitative, focusing on the design, development, and evaluation of the core prediction engine, formalized as the Crossmodal Temporal Fusion (CMTF) architecture. This phase initiates with comprehensive data collection and preprocessing, aligning historical market value data with localized Vietnamese financial news context. To establish robust performance benchmarks, an array of baseline models is developed for time-series market forecasting, including traditional Long Short-Term Memory (LSTM) networks, Random Forests, and modern foundational time-series models such as Chronos (evaluating both zero-shot and fine-tuned configurations) [21]. Concurrently, for the textual modality, PhoBERT [8] is fine-tuned specifically for financial sentiment analysis and news context embedding, and its efficacy is empirically compared against custom embedding strategies. The culmination of this phase involves engineering the CMTF to optimally fuse these distinct data modalities, followed by a rigorous statistical evaluation to compare the CMTF's predictive superiority against the established univariate and text-only baseline models.

Following the validation of the CMTF model, the second and third phases transition the research from isolated algorithmic prediction to robust systems engineering and user

interaction. The second phase upgrades the core prediction engine into a sophisticated multi-agent pipeline. Recognizing the inherent risks of standalone machine learning predictions in finance, this phase introduces deterministic critic agents designed to heuristically evaluate and dynamically adjust the CMTF’s initial outputs [37]. The efficacy of this system is evaluated quantitatively via an A/B testing framework, directly comparing the CMTF-only strategy against the multi-agent refined strategy. Furthermore, a realistic backtesting layer—incorporating trading costs, slippage, and continuous equity tracking—is implemented to validate the framework’s practical financial viability. Finally, the third phase translates the validated back-end architecture into a seamless User Interface (UI). This involves upgrading the multi-agent backend into a fully functional chatbot service, providing Vietnamese users with an intuitive, conversational gateway to interact with complex multimodal financial predictions.

Core Prediction Engine

The core prediction engine serves as the central analytical component of the proposed framework, designed to forecast financial asset prices by capturing the complex, multidimensional nature of market dynamics. Traditional forecasting models frequently rely on a single data modality, creating an inherent blind spot in highly volatile environments. To address this limitation, the engine is constructed around a multimodal architecture that simultaneously processes and fuses two distinct but deeply interconnected streams of information: quantitative time-series market data and qualitative textual signals.

Quantitative Time-Series Modality

The first foundational pillar of the prediction engine is derived from historical market mechanics. Financial markets generate high-frequency, structured numerical data that reflects the aggregate actions of market participants. The engine continuously ingests time-series signals, which include direct price history (open, high, low, close), trading volume, momentum indicators, and volatility metrics. These continuous signals are highly effective at identifying historical trends, cyclical market behaviors, and established support or resistance levels. However, time-series data is predominantly backward-looking; it reflects the outcome of market events rather than the catalysts driving them.

Qualitative Textual Modality

To capture the catalysts of market movement, the engine integrates a second, unstructured data modality consisting of textual signals. This stream encompasses financial news articles, official company reports, earnings call transcripts, and broader macroeconomic an-

nouncements. Textual data contains forward-looking sentiment, abrupt contextual shifts, and fundamental corporate changes that have not yet been fully priced into the historical trading data. By utilizing advanced language processing techniques, the engine extracts semantic meaning and sentiment polarity from these unstructured sources, converting subjective text into quantifiable predictive features.

Problem Formulation

The core prediction problem is formulated as a *supervised regression task over forward log-returns*, a representation that has been widely adopted in quantitative finance due to its superior mathematical and statistical properties compared to raw prices or simple percentage returns. For a given stock symbol and prediction cutoff time t , the model aims to estimate the future price movement over a horizon h via:

$$y_{t,h} = \log \left(\frac{P_{t+h}}{P_t} \right) \quad (1)$$

where P_t is the closing price at time t , and $h \in \{1, 5, 20\}$ trading days corresponds to short-, medium-, and long-term horizons.

Rationale for Log-Return as Target Variable

The use of log-return instead of raw price or arithmetic return is supported by both theoretical and empirical considerations.

First, log-returns exhibit *additivity and symmetry*. Unlike percentage returns, which suffer from asymmetric gain-loss scaling, log-returns preserve reversibility. For example, a price drop from 100 to 50 USD corresponds to a return of -50% , but recovering to 100 USD requires a $+100\%$ increase. In contrast, log-returns yield $\log(0.5) \approx -0.6931$ and $\log(2) \approx 0.6931$, whose sum equals zero, correctly reflecting a break-even condition [2].

Second, log-returns are approximately *stationary*, a crucial assumption for most statistical learning methods. Empirical studies using Augmented Dickey-Fuller (ADF) tests consistently show that while raw price series fail unit-root tests, log-return series strongly reject the null hypothesis of non-stationarity (i.e., very low p-values), thereby improving model stability and generalization.

Third, log-returns significantly enhance predictive performance in machine learning models. In a comparative experiment, predicting raw prices (Paradigm 1) resulted in poor performance, with Random Forest yielding $R^2 = -2.5950$ and LSTM suffering from a high RMSE of 187.83. When reformulated to predict log-returns (Paradigm 2), performance improved dramatically: LSTM achieved $R^2 = 0.8530$ with RMSE reduced to 42.98, while Random Forest reached $R^2 = 0.8382$ [38].

Additional empirical evidence across asset classes further supports this design choice. For example, in gold price prediction (2021–2025), a simple linear regression model achieved an almost perfect fit ($R^2 = 0.986$, $\text{RMSE} \approx 35.7$) when operating on transformed (stationary) data [39]. In high-frequency crude oil forecasting, hybrid APGARCH-ANN models leveraging log-return dynamics reduced RMSE by over 80% compared to classical parametric models [40]. Similarly, in cryptocurrency markets, hybrid GARCH-LSTM models achieved test RMSE as low as 0.011456, while Bi-LSTM models reported MAPE of only 1.16% on Ethereum price prediction tasks [41, 42].

Input Representation and Multimodal Structure

To capture both temporal dependencies and exogenous information, the model operates on a rolling look-back window of $T = 30$ trading days. The input is decomposed into two aligned tensors:

$$\mathbf{X}_t^m \in \mathbb{R}^{T \times d_m}, \quad \mathbf{X}_t^n \in \mathbb{R}^{T \times d_n} \quad (2)$$

where $\mathbf{X}_t^m$ represents market features ($d_m = 23$), including OHLCV variables and technical indicators, and $\mathbf{X}_t^n$ represents news features ($d_n = 773$), consisting of a 768-dimensional semantic embedding combined with 5 sentiment-derived scalars.

The learning objective is to approximate a horizon-specific function:

$$f_h : (\mathbf{X}_t^m, \mathbf{X}_t^n) \mapsto \hat{y}_{t,h} \quad (3)$$

where a separate model is trained for each horizon h . This design aligns with empirical findings that short-term returns are dominated by noise and momentum, while longer-term returns reflect macroeconomic and fundamental signals, making a shared parameterization suboptimal [2].

Practical Constraints and Known Limitations

Despite its advantages, the log-return formulation introduces several technical challenges. One key issue is the *Jensen's inequality bias*: transforming predicted log-returns back to price space via exponentiation leads to systematic underestimation. This bias is typically corrected by multiplying by $\exp(0.5\sigma^2)$, where σ^2 is the residual variance.

Another limitation arises in extreme market conditions. Since the logarithm is undefined for negative prices, events such as the WTI crude oil crash on April 20, 2020 (with prices reaching -37.63 USD/barrel) break the standard formulation. A common workaround is the *shifted log-return*, where a constant offset C is added to ensure positivity before transformation.

Information Constraint: No Look-Ahead Leakage

A strict no-look-ahead constraint is enforced across the entire pipeline. At time t , only information available up to t can be used for feature construction. This applies to market data, news alignment, and dataset splitting. Even subtle violations of this constraint can lead to inflated performance estimates and invalidate real-world applicability.

Auxiliary Sentiment Learning Task

Finally, the formulation includes an auxiliary supervised learning problem for sentiment extraction. Each news article is mapped to a probability distribution over sentiment classes (negative, neutral, positive), which is then converted into a scalar score $s(x) \in [-1, 1]$. This module is trained independently and serves as a feature generator for the downstream regression task, ensuring modularity and stability of the overall system.

Data Sources and Temporal Alignment Strategy

The construction of a reliable multimodal dataset for Vietnamese banking stocks presents challenges that are absent in mainstream financial forecasting benchmarks. Vietnamese financial markets are characterized by relatively limited data availability, language-specific news sources that are not covered by international aggregators, and irregular news publication patterns that demand careful temporal handling. This subsection describes the data collection scope and, more importantly, the methodological rationale for the temporal alignment strategy that governs how news events are mapped to market bars.

Market Data

Historical daily OHLCV data for the two target symbols—Vietcombank (VCB) and BIDV (BID)—is collected via the *vnstock* library, which provides access to the Ho Chi Minh Stock Exchange (HOSE). The study period spans from January 2022 to March 2026, providing approximately four years of daily observations. The choice of VCB and BID as the target universe is deliberate: both are among the most liquid and widely covered banking large-caps on the Vietnamese market, ensuring sufficient and consistent news coverage throughout the study period. Claiming broader generalization across all Vietnamese sectors or smaller-cap symbols is explicitly outside the scope of this research.

News Data

Vietnamese financial news is collected from three primary sources: *CafeF*, *VnExpress*, and *Vietstock*. These sources were selected because they collectively cover the primary dimensions of financial news relevant to banking stocks—capital market commentary,

macroeconomic policy announcements, corporate disclosures, and regulatory updates. The use of multiple sources reduces the risk that any single editorial bias or coverage gap systematically skews the news signal for a given period.

Temporal Alignment and Leakage Prevention

The most consequential methodological decision at the data preparation stage is how news articles are assigned to market bars. The naive approach—assigning any article dated on trading day T to bar T —introduces a subtle but severe form of data leakage. In practice, articles published after the market closes cannot have influenced the closing price of that same bar. Assigning such articles to bar T would therefore allow the model to exploit information that no market participant could have acted upon.

To eliminate this risk, this research adopts a *conservative 15:00 ICT cutoff rule*. Under this rule, an article published strictly before 15:00 Indochina Time on trading day T is considered available to bar T . Any article published at or after 15:00 is instead assigned to the next available trading bar. Articles published on weekends or public holidays are similarly shifted forward to the next trading session. The 15:00 threshold was selected because it precedes the typical HOSE closing time, ensuring that even articles published in the early afternoon are conservatively excluded from the same-day bar. This design intentionally sacrifices a marginal amount of same-day signal in exchange for a guarantee that the evaluation protocol remains valid and that reported model performance reflects realistic trading conditions rather than an optimistic backtest artifact [2].

Market Feature Engineering

Beyond raw OHLCV values, the market tensor is enriched with a set of standard technical indicators: RSI-14, MACD, Bollinger Bands, ATR-14, volume ratio, and daily log return. These indicators are not included because they are guaranteed to be optimal; rather, they encode established representations of market structure—momentum, mean-reversion tendency, volatility regime, and liquidity abnormality—that are economically meaningful dimensions in banking stocks and widely used in the quantitative finance literature [43, 44]. The engineering step therefore injects structured domain knowledge as an inductive bias, enabling the model to operate on interpretable market signals rather than purely on raw price values.

Vietnamese Sentiment Modeling Approach

Before news signals can be incorporated into the forecasting engine, a fundamental representational challenge must be resolved: Vietnamese financial text cannot be processed

by generic multilingual models without a meaningful loss of linguistic fidelity. Vietnamese is a tonal, analytic language with a writing system and vocabulary structure that differ substantially from the Indo-European languages that dominate most pretrained language model corpora. Financial terminology in Vietnamese further compounds this difficulty, as it includes domain-specific terms and abbreviations that are poorly represented even in general-purpose Vietnamese language models. This subsection describes the methodological approach adopted to overcome this challenge and produce a reusable, high-quality Vietnamese sentiment component.

Why a Dedicated Sentiment Stage

A two-stage architecture—in which sentiment modeling is performed separately and its outputs are subsequently consumed by the forecasting engine—is preferred over a fully end-to-end approach for three reasons. First, it separates the supervised sentiment training problem, which requires a labeled text dataset, from the forecasting problem, which requires historical market and news data. This separation prevents gradient signals from the forecasting objective from distorting the language model’s sentiment representations. Second, it produces a reusable artifact: the trained sentiment model can be exported and applied to any future news corpus without retraining the entire forecasting pipeline. Third, it allows independent evaluation of the text component, making it possible to verify that the sentiment module is actually learning meaningful Vietnamese financial sentiment before its outputs are injected into the downstream model.

Encoder Comparison Strategy

Two candidate title encoder architectures are evaluated under identical conditions to determine the most suitable backbone for Vietnamese financial sentiment. The first candidate is a *lightweight custom Transformer encoder*—a task-specific model trained from scratch on the target dataset, serving as a compact baseline that does not rely on large-scale pretraining. The second candidate is *PhoBERT* [8], a Vietnamese pretrained language model trained on a large-scale Vietnamese text corpus using the RoBERTa [45] framework. Both models are attached to the same classification head architecture so that the comparison isolates the contribution of the encoder backbone rather than confounding it with differences in the prediction head.

The methodological rationale for this comparison is to empirically answer a question that cannot be resolved by theory alone: is large-scale Vietnamese language pretraining necessary for adequate financial sentiment modeling, or is a compact task-trained model sufficient? If the PhoBERT branch demonstrates a clear advantage in terms of balanced classification quality, this justifies the additional computational cost of loading and fine-tuning

a pretrained language model for all downstream use. The final selection is governed by *macro-averaged F1* on the validation split, which equally weights precision and recall across all three sentiment classes and thereby prevents the classifier from exploiting class imbalance by over-predicting the dominant neutral class.

Sentiment Score as a Downstream Feature

The selected model produces not only class probabilities but also a scalar expected-value sentiment score $s(x) \in [-1, +1]$, computed as:

$$s(x) = \sum_{c \in \{-1, 0, +1\}} c \cdot p(c \mid x) \quad (4)$$

This continuous score serves dual purposes downstream. It acts as a directly interpretable market sentiment signal at article and bar level. It also provides the weighting coefficient for the sentiment-magnitude pooling of sentence embeddings, described in the following subsection. The export of the selected model as a frozen inference artifact—a *PhoBERT handoff bundle*—ensures that its outputs are stable and reproducible at all subsequent stages of the pipeline.

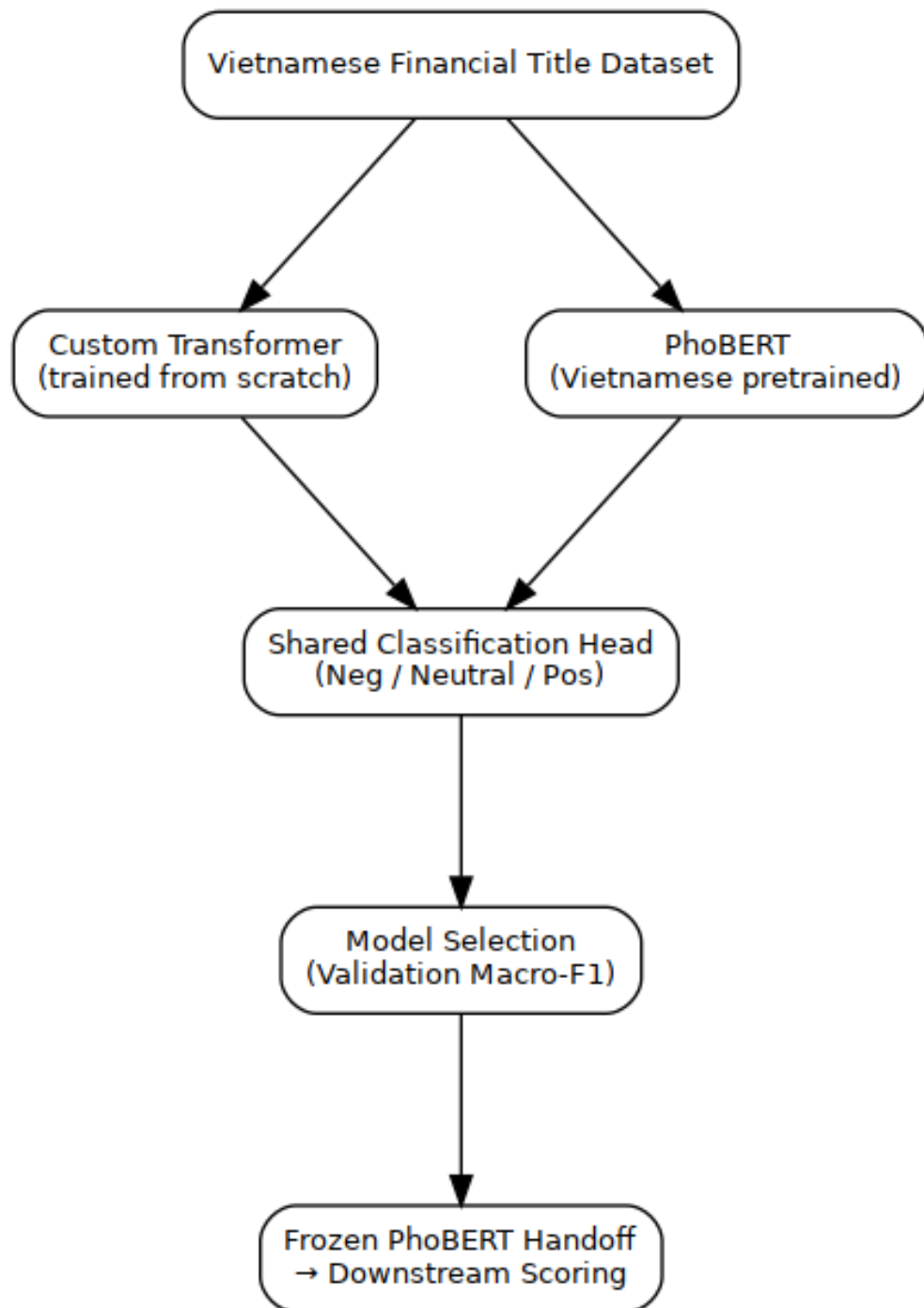


Figure 4.2: Vietnamese title sentiment modeling pipeline: parallel comparison of Custom Transformer and PhoBERT encoders feeding into the same classification head, with the selected PhoBERT checkpoint exported as a frozen handoff artifact.

Hybrid News Representation Design

Having established a reliable Vietnamese sentiment scorer, the next methodological challenge is converting a variable and often sparse collection of daily news articles into a fixed-dimensional, model-consumable representation for each market bar. This subsection describes the design of the *hybrid news vector*—a representation that deliberately combines semantic content with sentiment-derived statistics—and justifies each design decision.

Semantic Embedding Layer

Article text is encoded using `dangvantuan/vietnamese-embedding` [46], a Vietnamese sentence-transformer model that produces a 768-dimensional dense vector representing the semantic content of each article. The choice of a Vietnamese-specific sentence embedding model, rather than a generic multilingual encoder such as multilingual-BERT or LaBSE, is motivated by the same linguistic argument applied to the sentiment stage: Vietnamese-specific models exhibit better semantic alignment for Vietnamese text because they are trained on corpora where Vietnamese token distributions are dominant rather than diluted by hundreds of other languages. Sentence-transformer architectures are additionally preferred over token-level encoders for this pooling task because they are explicitly optimized for producing semantically meaningful sentence-level representations that support aggregation [47].

Sentiment-Magnitude Weighted Pooling

On any given trading day, a variable number of articles—ranging from zero to several—may be available for a given stock symbol. A naive uniform mean-pooling strategy treats all articles as equally informative, irrespective of their sentiment content. This is methodologically unsound in a financial context: a near-neutral article reporting routine administrative matters carries substantially less market-relevant information than a strongly negative article announcing a regulatory penalty or a strongly positive article announcing an earnings beat. To encode this asymmetry, the bar-level embedding is computed as a *sentiment-magnitude weighted average*:

$$\mathbf{e}_t = \frac{\sum_{i=1}^{n_t} |s_i| \mathbf{z}_i}{\sum_{i=1}^{n_t} |s_i| + \varepsilon} \quad (5)$$

where $\mathbf{z}_i \in \mathbb{R}^{768}$ is the sentence embedding of article i in bar t , s_i is its scalar sentiment score from the PhoBERT handoff, and ε is a small constant to prevent division by zero. The absolute

value of the sentiment score is used as the weight because both strongly positive and strongly negative articles are considered more market-relevant than neutral ones; polarity direction is already captured by the sign of s_i within the embedding and the scalar features. This pooling strategy is a computationally efficient heuristic that injects a market-relevance prior without requiring an additional learned attention mechanism at the article level.

Hybrid Vector Construction and Missing-News Handling

For each bar, the pooled 768-dimensional semantic vector is concatenated with five sentiment-derived scalar statistics—mean sentiment, maximum absolute sentiment, positive article ratio, negative article ratio, and article count—to form a 773-dimensional hybrid news vector. This concatenation is designed to preserve two complementary perspectives: the dense semantic vector captures the topical content and contextual nuance of the news, while the scalar statistics capture aggregate sentiment characteristics that may be predictive in their own right.

A sixth feature, the *sentiment missing flag*, is intentionally kept outside the news tensor and instead injected into the market feature pathway. This separation reflects a principled design decision: a zero news vector should not be semantically equivalent to a neutral news day. When no articles are available for a bar, the hybrid vector is set to all zeros, and the missing flag is set to one in the market feature pathway. This allows downstream models to distinguish between *absence of information*—which may itself be a signal in a news-rich environment—and *presence of neutral information*. Conflating the two would introduce a systematic representation error for all missing-news bars.

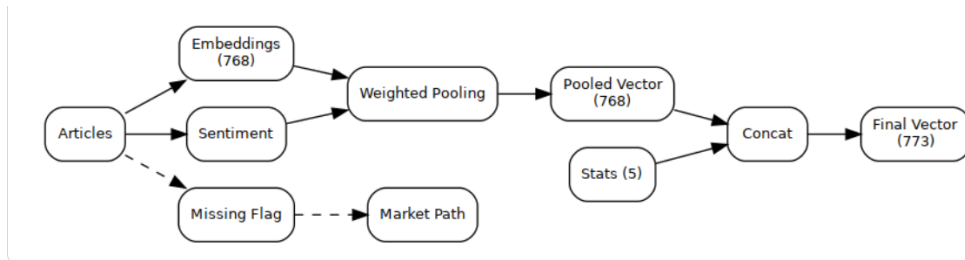


Figure 4.3: Construction of the 773-dimensional hybrid news vector per market bar: sentiment-weighted pooling of article embeddings concatenated with sentiment scalar features, with the missing-news flag routed separately into the market pathway.

Baseline Forecasting Models

A central principle of the proposed research methodology is that a novel forecasting architecture is only scientifically meaningful if its performance can be contextualized against

a set of well-chosen, diverse baselines evaluated under identical conditions. Introducing the Cross-Modal Temporal Fusion model without first establishing what simpler approaches can achieve would make it impossible to attribute any observed improvement to the specific architectural innovations of the CMTF rather than to incidental factors such as better data preprocessing or more favorable hyperparameter tuning. Four baseline model families are therefore included in the comparison, each chosen to isolate a distinct modeling question.

Random Forest Baseline

A Random Forest regressor operating over a flattened summary of the market window provides the *non-sequential tabular baseline*. This model does not assume any temporal ordering among features; it treats the market window as a fixed vector of engineered statistics. Its inclusion answers a fundamental prior question: how much predictive value is inherent in well-engineered market features alone, without any architectural capacity to model sequential dependencies? If a tree-based tabular model performs competitively with more complex sequence architectures, it would suggest that the additional inductive bias of recurrent or attention-based models is not warranted for this domain [48].

LSTM Baseline

A Long Short-Term Memory network operating directly over the 30-bar market window provides the *sequential recurrent baseline*. Unlike the Random Forest, the LSTM explicitly models the temporal order of market observations, enabling it to capture momentum, mean-reversion, and other path-dependent behaviors that are invisible to a tabular learner. The LSTM is a well-established benchmark in financial time-series forecasting and represents the strongest conventional deep-learning baseline before considering pretrained models [24].

Chronos Zero-Shot Baseline

The prediction task in this thesis is constrained by a small local sample, strong non-stationarity, and strict chronological validation with leakage control. Under this regime, fully in-domain deep sequence training is statistically fragile because parameter learning is exposed to high noise and limited regime diversity. Chronos [21] is therefore selected as a transfer-first baseline to import temporal priors learned from broad cross-domain pretraining before any market-specific adaptation is introduced.

From a theoretical perspective, Chronos inherits the Transformer design space of T5, where temporal dependence is modeled through self-attention over tokenized sequence context rather than recurrent state recursion. This architectural property is methodologically relevant for the present problem. First, long-range interactions can be represented without

the same memory bottlenecks that recurrent models face when horizon structure changes across windows. Second, the pretrained parameter space encodes reusable temporal regularities, including trend continuation, volatility shifts, and scale transitions, which are difficult to estimate robustly from Vietnamese banking data alone. In comparative terms, this places Chronos in a distinct position relative to other baselines: Random Forest does not model sequence order, while LSTM models order but relies exclusively on local data for representation learning.

In implementation, the zero-shot Chronos baseline is executed without domain parameter updates and is applied to the same leakage-safe walk-forward split used by all models. Its methodological role is to quantify pure transfer capability. Any performance gap between this configuration and adapted variants can then be interpreted as the measurable value of in-domain calibration rather than an artifact of evaluation inconsistency.

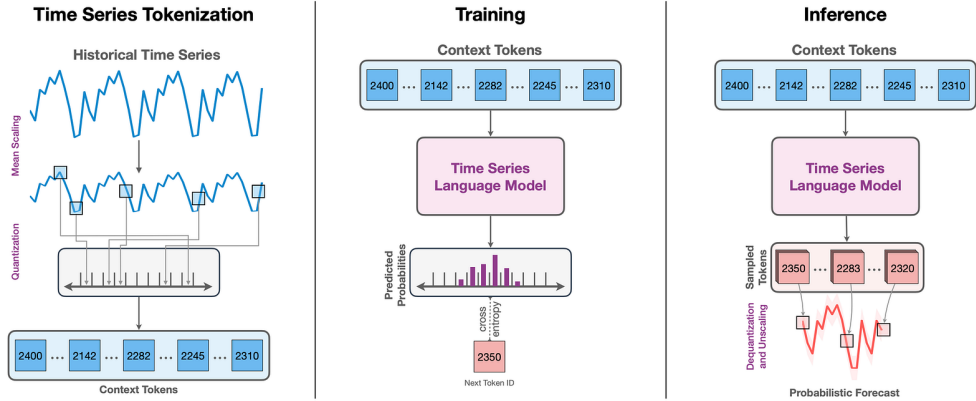


Figure 4.4: Example of the application of Uniform Binning

Chronos with LoRA Fine-Tuning

Although zero-shot transfer is informative, the target market remains distributionally distinct from generic pretraining corpora in terms of liquidity profile, event sensitivity, and regime transitions. Adaptation is therefore necessary, but unconstrained full fine-tuning is not preferred for this project because parameter volume is large relative to effective sample size. In this setting, full tuning risks absorbing idiosyncratic short-horizon noise and degrading generalization across evaluation periods.

The adaptation theory follows *Low-Rank Adaptation* (LoRA) [49]. Instead of updating full weight matrices, LoRA constrains updates to a low-rank subspace through factorized trainable matrices while the pretrained backbone remains largely frozen. This yields a regularized adaptation mechanism with direct relevance to the thesis constraints: reduced optimization variance, improved computational efficiency, and stronger retention of pretrained

temporal priors. Conceptually, LoRA allows the model to learn domain correction directions rather than relearn an entirely new temporal representation from limited Vietnamese data.

In implementation, LoRA adapters are injected into the Chronos attention pathway and trained under the same chronological split protocol as other baselines. The adapted Chronos branch is combined with a recurrent multivariate market branch and a current-bar tabular projection to form a richer market-state representation. This configuration is intentionally designated as the *strongest market-only baseline*. The methodological consequence is important for attribution: when CMTF is evaluated against this backbone under identical data and validation rules, improvements are interpretable as incremental multimodal contribution from news fusion rather than gains from unequal market-model capacity.

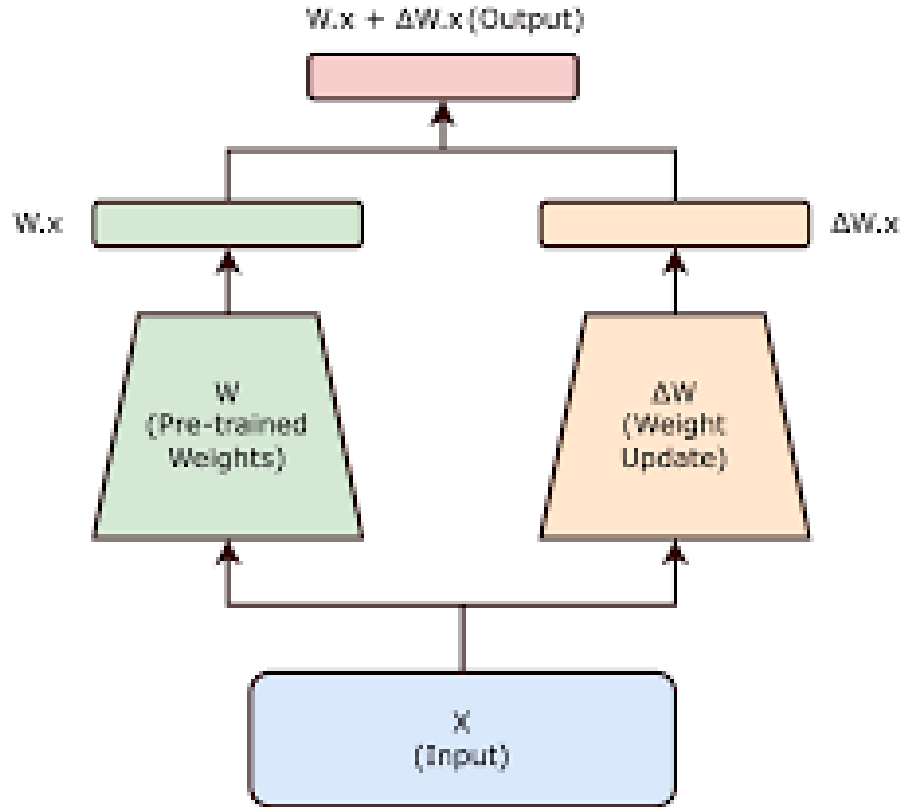


Figure 4.5: Conceptual LoRA placeholder showing low-rank adapter updates over frozen pretrained weights

Proposed Cross-Modal Temporal Fusion Architecture

The Cross-Modal Temporal Fusion (CMTF) model is the principal methodological contribution of this research phase. It is designed to answer the question that the baseline comparison leaves open: *does the systematic fusion of Vietnamese news signals with a strong market backbone produce measurably better forecasts than the market backbone alone?* The

architecture is built around a residual fusion principle that is both methodologically rigorous and practically robust, particularly in the context of the sparse and irregular news availability characteristic of the Vietnamese market.

Core Residual Design Principle

The CMTF prediction is formulated as a *baseline correction* rather than an end-to-end multimodal prediction:

$$\hat{y} = \hat{y}_{\text{base}} + g \cdot \alpha \cdot \phi(\mathbf{h}_{\text{base}}, \mathbf{X}_t^n) \quad (6)$$

where $\hat{y}_{\text{base}}$ is the prediction produced by the fine-tuned Chronos LoRA market backbone, $\mathbf{h}_{\text{base}} \in \mathbb{R}^{599}$ is the combined market feature vector from the backbone, $\phi(\cdot)$ is the cross-modal fusion head that estimates a news-conditioned residual, α is a learnable scalar parameter initialized conservatively that controls the global contribution of the news branch, and $g \in \{0, 1\}$ is a *zero-news gate* that is set to zero when no valid news is present in the entire 30-bar window.

This residual formulation provides three properties that a naive concatenation-based fusion cannot guarantee. First, *interpretability*: the magnitude of the news residual directly quantifies how much the news shifted the market forecast, making the model’s behavior auditable and explainable. Second, *zero-news parity*: when no news is available, $g = 0$ and the CMTF prediction is numerically identical to the market baseline, ensuring that the model does not introduce spurious behavior in the frequent case of missing news. Third, *stable initialization*: by starting with a conservative α and a well-initialized residual head, the model begins training close to the market-only solution and increases the influence of news only to the extent that the validation data supports it, preventing the fusion branch from corrupting a well-trained backbone in the early training epochs.

Cross-Attention Fusion Mechanism

The core of the fusion head $\phi(\cdot)$ is a *cross-attention* mechanism in which the market state serves as the query and the 30-bar sequence of hybrid news vectors serves as the key-value pairs [12]. This design is chosen over simpler alternatives—such as uniform mean-pooling of the news window or direct concatenation of the market and news tensors—because the news window is ordered, sparse, and temporally uneven. Not all 30 bars in the window carry equally relevant news: a major regulatory announcement three days ago should receive more attention than routine daily commentary from three weeks ago. Cross-attention gives the model a mechanism to learn this bar-level relevance directly from data, conditioned on the current market state. The market-conditioned query further ensures that relevance assessment

is context-sensitive: the same piece of news may be more or less important depending on what the market has been doing.

To enable the model to learn recency preferences, *learnable positional encodings* are added to the projected news sequence before the attention computation. Unlike fixed sinusoidal encodings, learnable positional embeddings allow the model to discover from the training data whether recent news bars are systematically more predictive than older ones, without imposing this as a hard prior.

Ensemble Inference Strategy

Rather than relying on a single trained instance, the CMTF is trained three times with different random seeds and the final prediction is computed as the mean of the three individual predictions. Ensemble averaging reduces variance arising from random initialization and the stochastic aspects of mini-batch training, both of which are amplified on relatively small financial datasets. The ensemble also provides a simple measure of prediction confidence: high disagreement across seeds on a particular sample indicates a less reliable forecast.

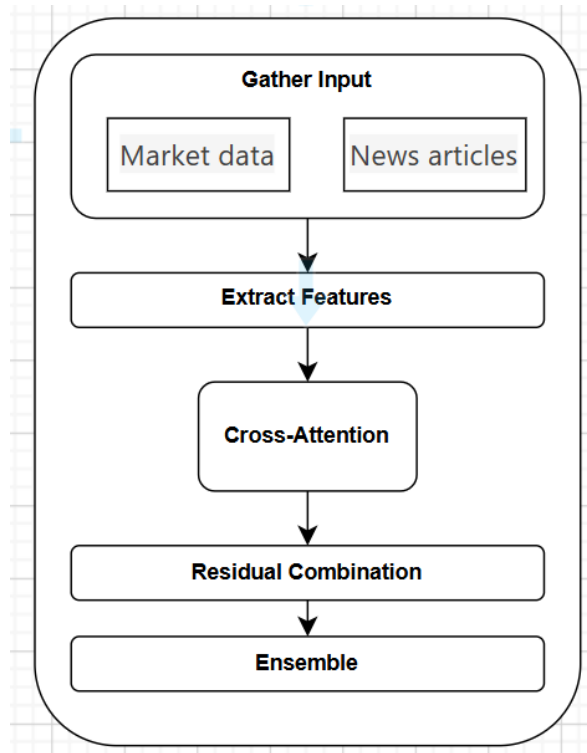


Figure 4.6: Proposed CMTF architecture: three market branches merging into a 599-dimensional market state, cross-modal fusion head with learnable positional encoding and multi-head cross-attention over the 30-bar news tensor, and residual combination with zero-news gate and learnable news weight.

Evaluation Design

The methodological value of the proposed prediction engine depends entirely on the rigor of its evaluation protocol. An evaluation design that inadvertently allows future information to leak into the training set, or that applies different preprocessing rules to different models, would produce results that are internally inconsistent and externally misleading. This subsection describes the evaluation strategy adopted to ensure that model comparisons are fair, temporally valid, and economically meaningful.

Walk-Forward Split with Horizon-Aware Purge

Financial time-series data is non-stationary: market regimes shift, volatility clusters, and correlations change over time. A random train-test split applied to such data inevitably mixes future observations into the training set and past observations into the test set, violating the temporal ordering that governs real-world prediction. To avoid this, the dataset is partitioned using a *walk-forward chronological split*: all samples with a prediction cutoff before June 30, 2024 form the training set; cutoffs between July 1, 2024 and December 31, 2024 form the validation set; and cutoffs after December 31, 2024 form the test set.

At each split boundary, a *horizon-aware purge* is applied: any training sample whose forward-return target overlaps with the validation period is removed, and analogously at the validation-test boundary. This purge prevents the model from indirectly learning the validation or test return through a training sample whose target window extends across the boundary [2]. All models in the benchmark are evaluated on the same cutoff lists, ensuring that observed performance differences reflect architectural choices rather than differences in evaluation data.

Multi-Dimensional Evaluation Metrics

No single metric adequately captures the usefulness of a financial forecasting model. A model can achieve low mean absolute error while still producing incorrect directional predictions that would result in consistent trading losses. Conversely, a model with high directional accuracy may produce predictions of such small magnitude that they are economically unactionable. To address this, the evaluation framework reports a comprehensive set of metrics spanning statistical accuracy and economic usefulness: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) for regression quality, Directional Accuracy (DA%) for sign prediction, Sharpe ratio and Information Coefficient (IC) for risk-adjusted economic relevance, and precision, recall, and F1 for directional classification views. A composite ranking score aggregates these dimensions to provide a single summary measure for model comparison, while the individual metrics remain available for detailed diagnostic analysis.

Success Criteria

The evaluation design defines success at two levels. At the sentiment stage, the selected Vietnamese encoder must demonstrate a sufficiently high macro-F1 to justify its deployment as a frozen inference artifact; an encoder that fails to reliably distinguish positive from negative sentiment would produce noise rather than signal in the downstream forecasting model. At the forecasting stage, the CMTF must outperform or compete strongly with the fine-tuned Chronos LoRA baseline—the strongest market-only model—on the held-out test protocol. Meeting this criterion is necessary to justify the additional architectural complexity of the cross-modal fusion branch and to validate the premise that Vietnamese news signals carry incremental predictive value beyond what is already encoded in historical price and volume data.

CHAPTER 5 - Implementation

CHAPTER 5 - CONCLUSION AND FUTURE WORK

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